

TED (10) - 4067
(REVISION - 2010)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/
TECHNOLOGY
HOSPITAL ENGINEERING
(common for BM and MD)

MODEL QUESTION PAPER

[Time : 3 hours]

(Maximum marks : 100)

PART – A

Marks

- I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define bio engineering?
2. Define a substation?
3. Give 4 examples for refrigerants?
4. What do you mean by quality control?
5. What do you mean by ISO standards?

(5×2=10)

PART – B

- II. Answer any *five* of the following. Each question carries 6 marks.

1. Describe Double chamber autoclave?
2. Explain different waste disposal methods?
3. Explain operation of a substation in a hospital?
4. Mention different specifications of hospital wiring?
5. Explain rotary compressors?
6. Explain the principle of year round air conditioning system ?
7. Explain different needs of a hospital administration?

(5×6=30)

PART – C

(Answer *one* full question from each unit. Each question carries 15 marks)

UNIT - I

- III. (a) Explain the duties and responsibilities of a hospital engineer (8)
(b) Distinguish the design of various wards (7)

OR

- IV. (a) Draw and explain briefly organisation chart? (8)
(b) Explain the design of ICU? (7)

UNIT – II

- V. (a) State the importance of electrical power supply system in a hospital? (8)
(b) Explain the working of a protective relay? (7)

OR

- VI. (a) Explain about different components of substation in hospital? (8)
(b) Explain the working of UPS with diagram? (7)

UNIT – III

- VII. (a) Explain gas supply system in a hospital? (9)
(b) Write a short note on
i) Theatre lighting (3)

- ii) Operation table (3)
- OR
- VIII. (a) Illustrate the principle of refrigeration system? (6)
(b) Explain different types of air purity filters? (9)
- UNIT – IV
- IX. (a) Explain the importance of ISO certification? (8)
(b) Specify the guidelines for selecting and maintaining biomedical equipments? (7)
- OR
- X. (a) Describe the ordering, testing, acceptance and maintenance protocol? (7)
(b) Explain about costing and financial planning in a hospital? (8)